

Compression and Deployment

DATA 443/643 · Week 7, note 7.3

Week 7

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Ideas from Stanford CS224N 2025 L11 (prune / PEFT as deploy moves). Original slides; that course is not copied.

Lecture plan

- 1 Prune, quantize, distill
- 2 Affine INT8 with a zero-point
- 3 Speculative decoding
- 4 What to report

Key goal: bits and tokens/s versus fp16, on the same eval, not only perplexity.

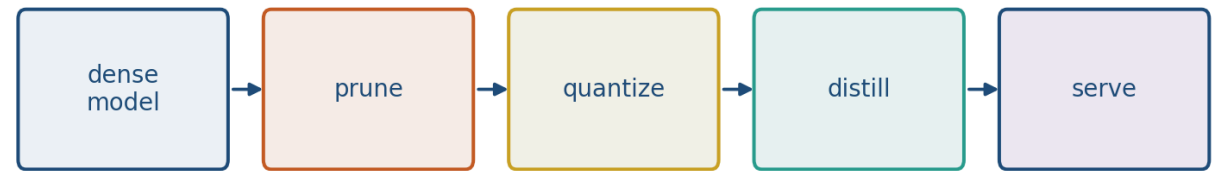
First time: four deployment methods

- What. Prune zeros weights. Quantize uses fewer bits. Distill trains a smaller student. Speculate drafts then verifies.
- Why. 7B in fp16 is ≈ 14 GB. Serving is memory and tokens/s, not only loss.
- How. Affine INT8: $x \approx s(q - z)$. Speculative accept if $p_{\text{target}}(x) \geq p_{\text{draft}}(x)$.
- Tradeoffs. + Fits RAM, faster decode. – Outliers blow the scale; a rejected draft wasted the small model.

A pipeline, not a slogan

- Prune: zeros or dropped heads.
Needs recovery training.
- Quantize: fewer bits. Distill: a smaller student.
- CS224N L11 also puts LoRA here.
Algebra is note 8.3.

Compression is a pipeline. Measure quality after each cut.



Scale and zero-point

$$x \approx s(q - z), \quad q \in \{0, \dots, 2^b - 1\}$$

On $[-1, 1]$, $b = 8$:
 $s = 2 / 255, z = 128$.

Drop z and you cannot represent negatives.

One activation outlier blows the scale.
Lab 7 shows the MSE jump.

Write the INT8 table

$s = 2/255$, $z = 128$. Drop z and the map cannot go below 0.

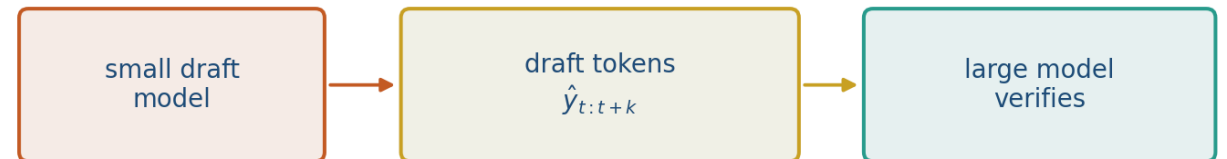
- $q = 128 \mapsto 0$. $q = 255 \mapsto \approx 0.996$.
- 7B fp16 is about 14 GB. INT4 weights about 3.5 GB plus overhead.
- Same architecture. Different bits.

INT8 on $[-1, 1]$	value
$q = 128$	$s(q - 128) = 0$
$q = 255$	$s \cdot 127 \approx 0.996$
no z	cannot represent negatives

Draft, then verify

- A cheap model proposes several tokens.
- The target checks them in one parallel pass.
- Reject at the first mismatch; resample from the target. The distribution does not change.

Speculative decoding: guess many tokens, accept the prefix the big model agrees on



What to take from this lecture

Need z

Centered maps still go negative.

Speculate $\neq$ distill

Extra small forwards to save large ones.

Report

Bits, layers, tokens/s, and the same downstream number.